

a Fresh requalification scope

384 fresh cases	49,152 formal samples	0 field-lineage overlap	5 held-out folds
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b All preregistered operational-identifiability criteria

NN median	P 0.0013 ≤ 0.250	P 8.04e-04 ≤ 0.250	N 0.277 ≤ 0.250	cell text: value frozen limit
NN P90	N 8.202 ≤ 0.600	N 45.54 ≤ 0.600	N 26.88 ≤ 0.600	
Cvar upper 95%	P 0.149 ≤ 0.350	N 1.070 ≤ 0.350	N 0.404 ≤ 0.350	
Oracle NRMSE	N 0.548 ≤ 0.500	N 1.010 ≤ 0.500	N 1.049 ≤ 0.500	
Oracle improvement	P 0.452 ≥ 0.200	N -0.0097 ≥ 0.200	N -0.049 ≥ 0.200	
Worst-family NRMSE	N 1.067 ≤ 0.750	N 1.004 ≤ 0.750	N 1.265 ≤ 0.750	
Coverage	P 0.936 ≥ 0.900	P 0.936 ≥ 0.900	P 0.936 ≥ 0.900	
	Density rate	Pressure gradient	Viscosity Laplacian	

Blue = criterion satisfied; orange = criterion not satisfied. Every row must pass for a component to qualify.

c Route conclusion

